



A retail customer being explained the smart meter installation process. A skilled technician can replace a single phase meter in 10 mins and without taking a power shutdown.

The Reform Framework

In 2021, the government approved the Revamped Distribution Sector Scheme (RDSS) with an outlay exceeding ₹3-lakh crore, according to the Union Ministry of Power. The scheme links financial support to measurable reform milestones, including loss reduction, improved billing efficiency and infrastructure modernisation.

A central instrument in that reform toolkit is the prepaid smart meter.

The national ambition is to install approximately 250 million prepaid smart meters over time. Unlike traditional electro-mechanical meters, smart meters transmit consumption data in real time, enable prepaid billing and support remote disconnection and reconnection. In principle, they eliminate estimated billing, reduce theft and tighten cash cycles for discoms. But policy ambition does not automatically translate into implementation. The scale of deployment required across India's diverse geography demands industrial execution capacity.

Execution as Differentiator

Adani Energy Solutions Limited (AESL), the transmission and distribution arm of the Adani Group portfolio, has emerged as one of the fastest-moving private players in this space. According to company disclosures, AESL has installed one crore smart meters, in approximately 25 months. Its current daily installation rate stands between 25,000 and

27,000 meters based on internal operational data. At this pace, the company expects to add another one crore meters within roughly 12 months.

Such speed is unusual in infrastructure. It reflects not only workforce mobilisation, but also process engineering. Smart metering contracts typically span 10 years or more. They include installation, communication infrastructure, software integration and long-term operations and maintenance. Payments are often structured as annuity-style revenues linked to performance metrics such as uptime and data integrity. In this model, execution quality directly influences margin sustainability.

The Financial Backbone

A defining feature of AESL's smart metering push is its financial architecture.

The company operates ~ 28,000 circuit kilometres of transmission lines across India, according to its latest investor disclosures. These assets function under Tariff-Based Competitive Bidding (TBCB) framework, regulated by the Central Electricity Regulatory Commission (CERC).

Revenues are contracted and predictable over the concession period, which typically extends over 30-35 years.

It also runs a profitable distribution business in Mumbai through subsidiary Adani Electricity Mumbai Limited (AEML). These businesses provide financial stability and balance sheet

strength to access long-term capital.

Smart metering, by nature, is capital intensive. Companies must procure meters, establish communication networks and deploy field teams before revenue streams stabilise.

Without balance sheet depth and cash flow predictability, rapid scaling in smart metering can strain financials. AESL's business model mitigates that risk by providing steady capital to support expansion.

From Manual to Systematised Execution

Infrastructure in India often falters not for lack of ambition, but due to fragmented execution. AESL has attempted to industrialise the rollout process, which has become its key differentiator vis-à-vis competitors.

A centralised Project Management Group coordinates planning, scheduling and quality governance across states. Standardised operating procedures replace decentralised contractor-led approaches. Digital workflow applications allocate tasks to field technicians, monitor geo-location compliance and confirm closure in real time. Artificial intelligence (AI)-enabled tools assist in quality checks, reducing installation errors and repeat visits. Senior leadership monitors performance through live dashboards. In high-volume deployment, incremental efficiency gains compound. A small improvement in first-time installation success rates can significantly lower lifecycle costs across millions of units.